

AI Usage Journal

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AI Tool Used: Gemini

Sections assisted: Quiz and Quick Reference

Section 1

I used Gemini Google, so that I can get ideas on what to create questions about which topics for the quiz and to help me identify common misconceptions people make for inspirations for the quiz.

Section 2

Prompt 1: “I am writing a multiple choice question about the limitations of block normalization in HOG. Based on the notes, can you list a few common misconceptions for me please”

Response: Gemini gave me the common misconception that block normalization has for example about extreme lighting and shadows.

Prompt 2: “Give me an idea for a trick question about HOG for my quiz”

Response: Gemini gave me an idea to make a question about how a single cell gets normalized multiple times.

Prompt 3: “Can you explain me what Hard Negative Mining is in my notes please as I don’t understand”

Response: Gemini provided an explanation about Hard Negative Mining and its definition

Prompt 4: “Can you give me ideas on what can I do for the feedback about each incorrect answer for my quiz as the maximum is 2 pages”

Response: Gemini gave me the idea of combining each incorrect feedback answer into 1 paragraph and shortening the explanation so it can fit into 2 pages

Section 3

1) Which concept in your topic did you find hardest to explain to someone else, and why do you think that was?

Response: The hardest concept to explain was HOG's overlapping blocks. It is difficult to explain why an 8x8 pixel cell is normalized multiple times by being part of four different 2x2 blocks and how this forms the secret sauce of the descriptor. Without visual diagrams, describing this spatial overlap and its impact on the final feature vector to someone that doesn't know anything about it is quite difficult.

2) Where did AI give you something that looked correct but turned out to be wrong or incomplete? What did you do about it?

Response: When I initially asked the AI to expand on why every individual incorrect option was wrong, it gave me a massive, overly detailed output. While it is correct, it was going to be much more than the Maximum 2 page limit and to counter this I asked it to give me ideas on what to do and as a result it gave me that I needed to make them into 1 paragraph and shorten the word.

3) Describe a specific moment where using AI wasted your time. What happened and what did you learn from it?

Response: It was when I made it to give me ideas about common misconceptions for the quiz but I wasn't clear enough about the 2 page limit. And from this I learned to always be clear and detailed in my prompts for it.

4) Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone. What made the difference?

Response: The AI was incredibly effective at generating the incorrect answer feedback for the multiple choice options. It expanded on why people might not know them and why a student might mistakenly choose the incorrect option and this saved a lot of time and energy.

5) What question about your topic does your team still not feel confident answering, even after building the entire Learning Pack?

Response: I think it has to be about the Sliding Window of Dalal and Triggs. It is about none standard feature vector calculations as even if it is explained very well in the notes, I would still struggle to write a script about it.

AI Usage Journal - HOG Learning Pack

Topic: Histogram of Oriented Gradients (HOG)

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Purpose. This journal records how AI was used during the creation of the HOG study notes. It focuses on specific uses, useful prompts, errors, wasted time, and what was learned rather than simply stating that AI was used.

Section 1: Tools Declared

Tool	What it was used for
ChatGPT (GPT-5.5 Thinking)	Used to explain HOG concepts, improve wording, structure study notes, generate worked examples, suggest citation placement, and draft this AI usage journal.
Python / python-docx via ChatGPT	Used through ChatGPT to generate Word documents from the written notes and journal drafts.
ChatGPT document rendering/checking workflow	Used to render generated DOCX files into page images so layout issues could be checked before final download.

Section 2: Notable Prompts

#	Prompt text	Brief summary of response / effect on work
1	"I need to understand everything about HOG... I have 5 hours... bombard me with information."	This started the learning process. The response gave a broad crash-course style explanation of HOG, including gradients, cells, blocks, normalisation, and the overall pipeline. It helped decide the main structure of the notes.
2	"I am making STUDY NOTES for this HOG project... Where do I start?"	AI generated the first draft of the notes in a structured Word document. This was useful as a starting point, but it was too short and had to be expanded significantly.
3	"Imagine you are giving this as material to people that know nothing about HOG but have a technical background... is it Informative enough?"	This prompt helped push the notes from a short summary toward a teaching resource. It led to more intuition, more mathematical background, pipeline diagrams, parameter choices, failure modes, and real-world applications.
4	"Take this and make it more technical..." followed by the gradient/cell/block explanation.	This improved the technical precision of the explanation. It helped replace simple wording with more accurate discussion of local intensity change, gradient orientation, magnitude weighting, cell histograms, and spatial structure.
5	"Walk me through a full numerical example + examiner insights."	This produced the numerical HOG examples used to explain Gx, Gy, magnitude, orientation, bin interpolation, cell histograms, and block normalisation. It made the notes more practical and exam-friendly.
6	"Where can I cite this?" and follow-up prompts about Viola-Jones, HOGgles, and Dalal-Triggs-Schmid.	These prompts helped place sources in the correct parts of the report. AI explained which papers supported historical context, HOG failure modes, and video/motion extensions, preventing citations from being used in irrelevant places.

Section 3: Five Questions

1. Which concept in your topic did you find hardest to explain to someone else, and why do you think that was?

The hardest concept to explain was block normalisation, especially why overlapping blocks improve robustness. Gradients and cells are quite intuitive because they connect to edges and local shape. Normalisation is harder because it is more abstract: the descriptor changes from raw edge strengths to relative edge patterns. I needed AI to explain it in several ways, including the dark-image versus bright-image example, before it became clear enough to teach.

2. Where did AI give you something that looked correct but turned out to be wrong or incomplete? What did you do about it?

AI initially produced material that looked polished but was incomplete for the assignment level. For example, the first notes were structurally correct but far too short and not detailed enough for a 12-page learning pack. It also produced video references that did not link properly. I responded by asking for deeper iterations, numerical examples, source alignment, and citation placement. I also checked whether each citation actually supported the claim being made.

3. Describe a specific moment where using AI wasted your time. What happened and what did you learn from it?

A clear wasted-time moment was when AI created a “final” HOG document too early. It sounded confident, but the document was not actually deep enough and did not match the requested length or teaching style. I had to redirect it several times with stricter requirements. I learned that I should not accept a polished AI output just because it sounds finished; I need to compare it against the marking criteria and ask for targeted revisions.

4. Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone. What made the difference?

AI was genuinely helpful when creating the worked numerical HOG example. I understood the general idea of gradients and bins, but AI helped turn it into a step-by-step teaching explanation with G_x , G_y , magnitude, orientation, vote splitting, histograms, and block normalisation. The difference was that AI could quickly reformat the same idea into clearer versions: technical, shorter, use LaTeX, table-based, and report-ready.

5. What question about your topic does your team still not feel confident answering, even after building the entire Learning Pack?

The remaining weak area is how HOG behaves in a full implementation under different parameter combinations on real datasets. We can explain why 8x8 cells, 9 bins, overlapping 2x2 blocks, and a 64x128 window are standard, but we are less confident predicting performance changes quantitatively. For example, we can describe why smaller cells increase noise, but not exactly how much accuracy or runtime would change without running experiments.

Reflection note. The AI was most useful as a tutor, formatter, and revision partner. It was weakest when treated as automatically correct. The final work improved when prompts became specific, when outputs were checked against source material, and when mistakes were openly corrected.

Section 1: Tools Declared

Claude AI (Sonnet 4.6): Used for the most part of the assignment. Helped generate the structure of the notebook.

Gemini 3: Used in cases where I needed errors to be explained in further detail to see what went wrong and how I might fix it.

Section 2: Notable prompts

AI used: Claude

Prompt: Help me write the annotated code walkthrough for Histogram of Oriented Gradients. Include gradient computation, cell histograms, block normalisation, full pipeline, parameter choices and failure modes.

Response: This created the base of the journal where most of the markdown and code cells were created, starting with the main imports and continuing to include descriptions of the markdown and code. It also included information about the mathematics behind

Prompt: Are certain values that are being used in the code forced to be like this? Can they be generalized or changed?

Response: This told me why the AI decided to use certain hardcoded values in those areas. (Example 8x8 pixel cell in cell histograms) and why each value was so important be how it was in respect to the code walkthrough

AI used: Gemini

Prompt: I got this error: “NameError: name 'cell_histograms' is not defined”. Why am I given this error and how do I fix this.

Response: The AI tried to tell me alterations that I could try and do to my code in order to make it work but none of these worked. It tried to define the histogram and its parameters, number of cells, histogram array and try to build it but that didn't work. Eventually, it told me to check the name of the image variable, and I realised it was incorrect and that's why the error was popping up.

Prompt: Please explain the significance of these graphs and why they are important. (Inserted the image of the feature vector after L2 block normalization and cell Histogram before normalization)

Response: Gave me information about the two graphs and the information I can extract from them. Feature vectors represent the structural signature of the image and how an AI classifier will recognise the shape regardless when the time of day was or how the lighting is. Cell

histograms represent the local geometry of the image but is flawed from illumination changes and shadowing.

Section 3: Five questions

1. The hardest part was explaining block normalization and how the cells overlap. It is confusing how cells group together to form bigger blocks, and since these blocks slide across the image and overlap, a single cell can be inside multiple blocks. Explaining how this helps fix problems with bad lighting and shadows is confusing to me.
2. This happened when in the original generation of the code walkthrough the AI gave me a graph that looked correct. When we looked closer, we realised that it had zero use and meaning to it as the x and y axis weren't correlated. I asked the AI and it told me that I was right and changed it to be something of use to show failure cases.
3. As mentioned above, Gemini tried to fix my code by complicating things when the answer was simple. It tried to add more details to the histogram but didn't realise that the variable name was incorrect. Using the code Gemini provided proved useless and I had to find out what's wrong myself. This made me learn that not all AI says can be accurate.
4. AI made my part faster as it helped give me the initial skeleton of the code walkthrough. It gave me a lot of good information and code cells that were kept the same in the final deliverable as well. Despite that there needed to be some changes that were obviously wrong, it gave me a lot of really good information.
5. I still would not feel comfortable answering deep questions about SVMs. While I completely understand how the HOG pipeline processes images, I am not sure how it pairs up with an SVM classifier in the real world. It is hard to visualize how a machine learning model takes that massive list of HOG numbers and uses them to spot a person.

AI Usage Journal - Eytan Attard Vella (0206606L)

Section 1: Tools declared

NotebookLM - Provided the basis of the learning regarding the theoretical foundation of HOG, with excellent insights from the papers we obtained (including the original paper).

Claude (Sonnet 4.6) - The main AI tool which helped to further understand the HOG method and developing the simulator and parts of the presentation.

Section 2: Notable Prompts

Prompt 1 (for answer 2): Can you explain bilinear interpolation in the context of HOG's orientation voting? When I check the definition of bilinear interpolation it seems like what the slide shows is just linear interpolation."

Summarised response 1: The orientation split alone is linear, but the complete implementation also interpolates spatially, making it genuinely bilinear.

Prompt 2 (for answer 4): "I am building a HOG simulator using Gradio and scikit-image. I need the HOG visualisation to update automatically whenever any of the four inputs change, the image upload, orientations slider, pixels per cell slider, and cells per block slider. I also want example images to be pre-cached so they load instantly."

Summarised response 2: Produced a clean Gradio Blocks structure with `.change()` event triggers chained across all four inputs sharing a single input list, and `cache_examples=True` applied to the `gr.Examples` block to pre-compute outputs at launch. The code was immediately runnable.

Prompt 3 (for answer 1): "Can you explain intuitively why overlapping blocks improve HOG's robustness? I understand the mathematics of L2 normalisation but I am struggling to explain why the same cell appearing in multiple blocks is useful rather than just redundant."

Summarised response 3: Each block normalises its cells relative to its own local neighbourhood. When a cell appears in multiple overlapping blocks, it gets rescaled against several different surrounding contexts. This means a localised lighting change or strong edge in one part of the block affects only that block's normalisation, not the cell's representation in every other block it belongs to. The redundancy is precisely what makes the descriptor more stable — the same cell's gradient pattern is preserved across different normalisation contexts rather than being dominated by one strong neighbour.

Prompt 4 (for answer 5): "Our Learning Pack covers the HOG descriptor pipeline in full (gradients, cells, blocks, normalisation, and feature vector construction). We did not really cover the SVM training side. Can you briefly explain how it works in a HOG plus SVM pedestrian detector and why it is necessary?"

Summarised response 4: A HOG + SVM detector is initially trained on positive examples (pedestrian windows) and randomly sampled negative examples (background patches). After this first round, the trained detector is run on the negative images and any false positives — background patches the detector incorrectly flagged as pedestrians — are collected and added to the training set as hard negatives. The classifier is then retrained. This process is repeated because random background patches are too easy; the classifier needs to see the specific gradient patterns that fool it in order to learn to reject them.

Section 3: Five Questions

1. Which concept in your topic did you find hardest to explain to someone else, and why do you think that was?

Certain technical aspects of HOG such as block normalisation was the hardest for me to explain, particularly why overlapping blocks are useful in the first place. The mathematics itself is simple, but the intuition of why the same cell getting normalised against neighbouring cells (and how that improves robustness rather than just adding redundancy) is required was a struggle to put into words.

2. Where did AI give you something that looked correct but turned out to be wrong or incomplete? What did you do about it?

Discussing the Python notebook implementation with my teammate made me realise that AI was constantly suggesting the use of functions which create simple images (such as stick figures, shapes, etc) rather than actual images. This would lead to poor analysis when reviewing details such as parameter choices, as the effect of the number of orientation bins was not being highlighted successfully, so by utilising proper images this helped bring out the intended insights (i.e. less bins = poor angle discrimination, more bins = precise but sparse). Thus proper images were utilised. Additionally, when working on the presentation slides, AI attempted to phrase things in certain ways which seemed misleading such as "Bilinear vote split" as the label for Step 3 of the worked example. Upon closer inspection and cross-referencing with the definition of bilinear interpolation, it became apparent that the operation shown on the slide (splitting a vote between two neighbouring orientation bins along a single axis) is technically linear interpolation, not bilinear. Bilinear interpolation requires two independent axes, and while the full HOG implementation does interpolate across both orientation and spatial dimensions simultaneously (making it genuinely bilinear in practice), the slide only demonstrated the orientation axis by itself. This was misleading to a teammate of mine who checked the definition separately. The label was discussed and a clearer understanding was reached in case anyone required an explanation.

3. Describe a specific moment where using AI wasted your time. What happened and what did you learn from it?

When working on the presentation slides, AI produced a worked example normalisation step where the block vector length was given as approximately 707.1. This figure came from a simplified assumption in the study notes, that all four cells in the block are identical and the sum of squares is exactly 500,000, which is a teaching approximation rather than a computed value. The number looked plausible and was consistent with the notes, so I did not immediately question it. It was only after checking with the corrected version of the deck that the discrepancy was noticed and the value was updated to approximately 510. Time was lost because the output appeared reasonable on the surface, and the error only surfaced through comparison rather than through catching it directly. The lesson learned was that numerically specific outputs from AI require verification, particularly when the numbers are carried over from a simplified example rather than freshly derived.

4. Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone. What made the difference?

When building the simulator, AI was particularly useful in structuring the Gradio interface cleanly and efficiently. Specifically, setting up the event triggers so that the HOG visualisation updates automatically whenever any slider changes (not just when the image is uploaded), required chaining multiple `.change()` calls across all four inputs simultaneously. While this is not conceptually difficult,

getting the input list right and ensuring `cache_examples=True` was applied correctly to pre-compute the example outputs would have required considerable time against the Gradio documentation. AI produced a clean working structure immediately, which meant time could be spent on the conceptual decisions, such as choosing the slider ranges and design choices such as preloaded example images, rather than on boilerplate code. The difference was that AI handled the code efficiently, freeing the focus for the parts of the simulator that required genuine understanding of HOG theory (e.g. selecting appropriate example images).

5. What question about your topic does your team still not feel confident answering, even after building the entire Learning Pack?

The team does not feel confident answering questions about how HOG performs in practice when combined with a linear SVM classifier, specifically the training side of the pipeline. Throughout the Learning Pack, the focus was placed on the descriptor itself (how the feature vector is constructed, what each parameter controls, and where the representation breaks down). However, questions such as how many positive and negative training examples are typically needed and how hard negative mining works in detail are ones the team would struggle to answer. This felt like where HOG stops being a standalone concept and becomes part of a broader machine learning pipeline that was not fully explored.

AI Use Journal

HOG Feature Descriptor — Learning Pack

Section 1: Tools Declared

The following AI tools were used in the production of this Learning Pack:

Tool	Primary Use
ChatGPT	Restructuring, rewording, and reducing word count of draft notes; generating an outline/plan structure for the Learning Pack.
Gemini Thinking & Pro	Attached to our NotebookLM source collection to enable source-grounded prompting;
Claude	Attempted visualisation generation for HOG pipeline diagrams; visualisations were ultimately not used in the final submission.

Section 2: Notable Prompts

Prompt 1

"Explain how the window and block differ within the context of HOG."

Response Summary:

The model explained that blocks serve a normalisation function, grouping cells together to improve robustness against illumination changes while windows slide across the image to capture a distinct descriptor over different local regions. It also clarified what happens when block and window sizes are increased or decreased. This resolved a genuine confusion that had been slowing down the writing process.

Prompt 2

"Can you provide me with some academic sources that are best to refer to when writing a study pack about HOG?"

Response Summary:

The model offered several recommendations, including the foundational Dalal & Triggs (2005) paper, which was legitimate and useful. However, it also generated several sources that either did not exist or had invalid links. This was identified by attempting to verify each reference, which was time-consuming and ultimately unsuccessful for multiple suggested sources.

Prompt 3

"Can you please improve the following section from my notes which explains [concept]? Focus on clarity and explaining the concepts in a clear and simple way."

Response Summary:

The model reliably rewrote sections in a way that preserved the original meaning and intent while improving clarity and conciseness. It did not introduce new claims or alter the key ideas, it simply tightened the language. This was one of the most consistently useful applications of AI throughout the project.

Prompt 4

"Can you explain all the key hyperparameters of HOG? As well explain the effects of different adjustments you could make to these hyperparameters."

Response Summary:

The model produced a well-structured summary of HOG hyperparameters, including cell size, block size, number of orientation bins, and stride, and explained the effect of increasing or decreasing each. This served as a useful reference when expanding the corresponding section of the Learning Pack, though it still required cross-checking against established sources.

Section 3: Five Questions

Q1. Which concept in your topic did you find hardest to explain to someone else, and why do you think that was?

The distinction between blocks and windows within the HOG pipeline was the most difficult concept to communicate clearly. Both involve a sliding mechanism, and that surface-level similarity made it easy to conflate the two, even after individually understanding each. The confusion stemmed from not having a precise mental model of the ordering and purpose of each level of the hierarchy. Once the respective roles (normalisation for blocks; descriptor capture for windows) were framed clearly, the distinction became much easier to articulate.

Q2. Where did AI give you something that looked correct but turned out to be wrong or incomplete? What did you do about it?

ChatGPT was unreliable when asked to recommend academic sources. It generated plausible-sounding paper titles and author combinations, some of which appeared credible at first glance, including references that resembled legitimate papers. Attempting to locate them via Google Scholar and the provided links revealed they either did not exist or had broken URLs. The response was to abandon AI-assisted source finding entirely and locate references manually, which proved faster and more reliable than iterating on AI-generated lists.

Q3. Describe a specific moment where using AI wasted your time. What happened and what did you learn from it?

Spending time prompting ChatGPT for academic sources, then verifying each one individually, cost more time than a direct literature search would have. Several sources recommended by the AI were entirely fabricated; others existed but with incorrect metadata or dead links. The lesson was clear: AI tools are poor at tasks requiring factual accuracy about external artefacts specifically, bibliographic data. Going forward, source-finding was done manually from the outset, which was both faster and produced higher-quality references.

Q4. Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone. What made the difference?

The most consistent value came from using AI to convert rough bullet-point notes into coherent prose. After drafting the core ideas, prompting the model to improve clarity and reduce word count, without changing the substance, produced polished paragraphs significantly faster than writing from scratch. The key factor was that the conceptual content was already correct; the model was only improving expression. When AI was used for generation rather than refinement, the results were far less reliable.

Q5. What question about your topic does your team still not feel confident answering, even after building the entire Learning Pack?

The interactions between multiple hyperparameters remain underexplored. While the effect of adjusting any single hyperparameter, cell size, block size, orientation bins, stride, is reasonably well understood in isolation, predicting how simultaneous changes across several of these interact is less clear. For example, how a reduction in cell size combined with a wider block stride affects overall descriptor quality in varying image conditions is not something the team could answer with confidence without empirical testing.